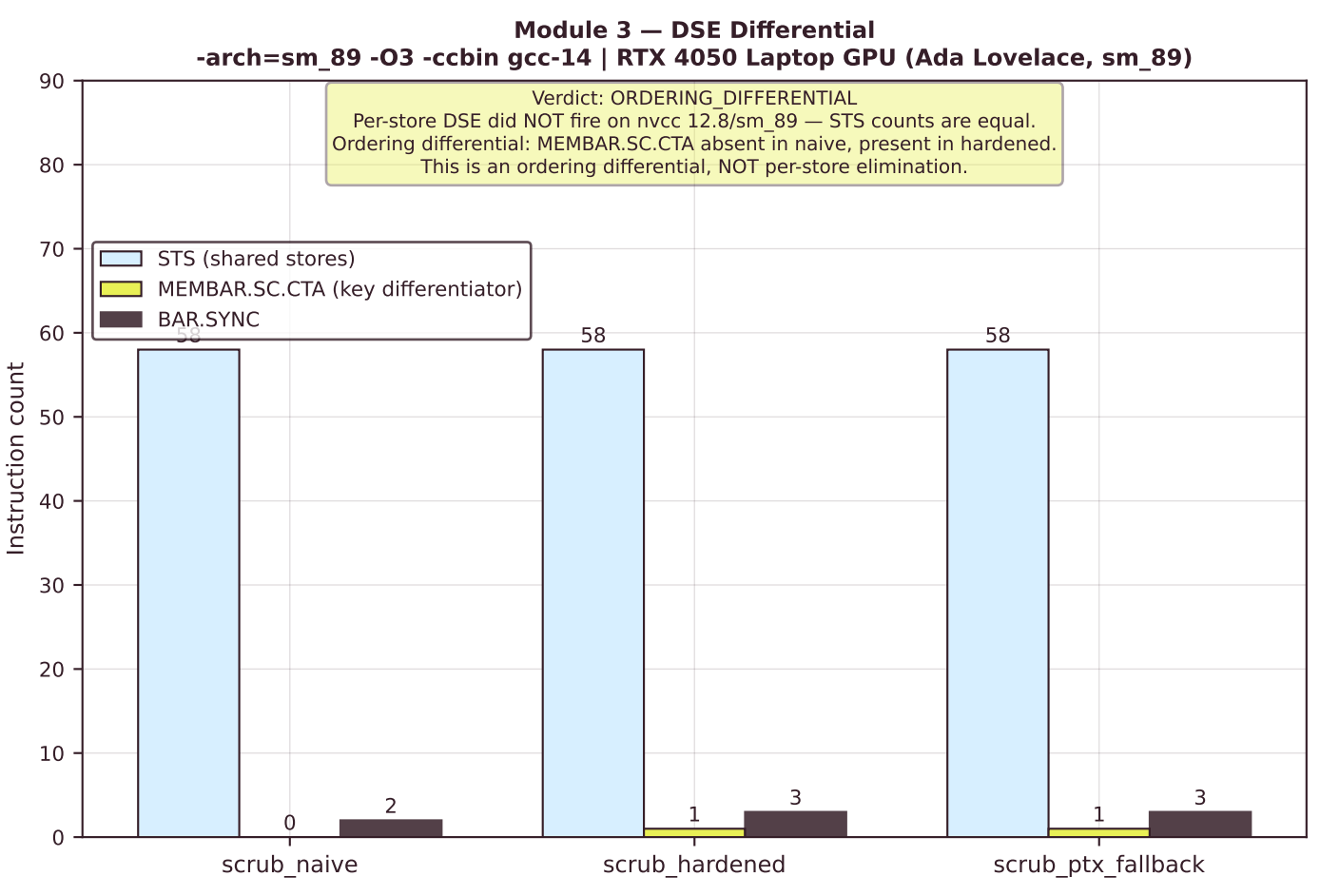


Annotated SASS excerpt

```
── scrub_naive (no ordering guarantee) ──────
  STS R0, [R2+0x000]
  STS R0, [R2+0x004]
  ...  (58 STS total)
  BAR.SYNC 0      ← __syncthreads
                    (no MEMBAR before)

── scrub_hardened (LMHC pattern) ────────────
  STS R0, [R2+0x000]
  STS R0, [R2+0x004]
  ...  (58 STS total)
  MEMBAR.SC.CTA   ← __threadfence_block
                    (C++ happens-before)
  BAR.SYNC 0      ← __syncthreads

STS counts equal → DSE did not fire.
MEMBAR absent in naive → no ordering
guarantee at kernel exit (Theorem 1).
```